

Qualification Pack



Technician - Power Substation Erection and Commissioning

QP Code: PSS/Q2502

Version: 1.0

NSQF Level: 4

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PSS/Q2502: Technician - Power Substation Erection and Commissioning

Brief Job Description

Technician-Substation Erection and Commissioning is responsible for carrying out all types of technical activities for construction of a new substation (66/11kV and 33/11 kV) for erection and commissioning, which include installation of various equipment in the control room and outdoor switchyard under the supervision of Jr. Engineer (Executive). The substation Technician installs and connects all indoor and outdoor equipment as per schematic lay out plan of grid station. The person carries out control cable wiring and circuit from yard to indoor control panel.

Personal Attributes

The individual should have the ability to read, write and follow the instructions of officer in-charge and possess the ability to stand for long working hours. The individual should also be physically fit and have an eye for detail.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1331: Apply basic health and safety practices for power related work](#)
2. [PSS/N1336: Working effectively with others](#)
3. [PSS/N2503: Prepare for primary and secondary system of grid substation erection](#)
4. [PSS/N2504: Erect and install substation equipment](#)
5. [PSS/N2505: Perform testing, commissioning and energization of substation equipment](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
Country	India
NSQF Level	4



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Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3113
Minimum Educational Qualification & Experience	I.T.I (Electrician) with 2-3 Years of experience Assistant Substation Technician/Apprentice
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	02/05/2019
Next Review Date	02/05/2023
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03479
NQR Version	1.0



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PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Scope

This unit/task covers the following:

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** • use protective clothing/equipment for specific tasks and work conditions
Protective Clothing
• leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visor
equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace
Hazards- electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards
possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions
- PC3.** follow procedures for safe working with electrical equipment
Safe Working Procedures: tag out/lock out, ptw (permit to work)
- PC4.** follow warning signs while working with electrical systems
Warning Signs: danger, high voltage area, out of service, etc
- PC5.** use standard safe working practices when working at heights, confined areas and trenches
- PC6.** test any electrical equipment and system using insulated testing devices before touching them
- PC7.** ensure positive isolation of electrical equipment & system as per the given standards
- PC8.** identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installed
Parameters: temperature, pressure, flow and current

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- PC9.** follow safe working practices while working in hazardous conditions to ensure the safety of self and others
Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.
- PC10.** use various methods of accident prevention in the work environment
Methods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors
- PC11.** identify and retrieve general health and safety equipment in the workplace
General Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.
- PC12.** set up and use scaffolds, elevated platforms and ladders safely
Set up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc
- PC13.** inspect scaffolds, elevated platforms and ladders for faults and rectify them
Faults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.
- PC14.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa
- PC15.** inspect power plant and its equipment routinely for any signs of oil, water and steam leakage
- PC16.** store flammable materials and machine lubricating oil safely and correctly
- PC17.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC18.** use good housekeeping practices at all times
Good Housekeeping Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces
- PC19.** identify common hazard signs displayed in various areas
Various Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.
- PC20.** inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

- PC21.** use various appropriate fire extinguishers for different types of fires correctly
Types of Fires: class a, b, c, d and e
- PC22.** use appropriate rescue techniques during fire hazard
- PC23.** maintain good housekeeping in order to prevent fire hazards
- PC24.** use a fire extinguisher correctly in case of fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC25.** administer appropriate first aid to the victims for the relevant injuries
Injuries: bleeding, burns, choking, electric shock, poisoning etc.
- PC26.** act promptly and appropriately to an accident situation or medical emergency in real or simulated environments



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- PC27.** perform and organise loss minimization or rescue activity during an accident in real or simulated environments
- PC28.** administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services
- PC29.** demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergencies
Emergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work
- PC30.** free an affected person from electrocution safely as per the recommended procedure
- PC31.** move injured people and others to a safe place during emergencies
- PC32.** write an incident report or dictate a report to another person, and send report to the authorized person
Incident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
2. names and location of documents that refer to health and safety in the workplace
3. names and location of people responsible for health and safety in the workplace
4. documents that refer to health and safety in the workplace
documents: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (e.g. government notices)
5. meaning of hazards and risks
6. health and safety hazards commonly present in the work environment and related precautions
7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
8. possible causes of risk and accident
possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness); not taking safety precautions
9. methods of accident prevention
10. safe working practices when working with tools and machines
11. safe working practices while working at various hazardous sites
12. where to find all the general health and safety equipment in the workplace
13. various dangers associated with the use of electrical equipment
14. positive isolation of electrical equipment and system
15. safe handling and disposal of hazardous power plant wastes
16. use of emission and pollution control devices and measures taken to control pollution
17. various safety procedures and equipment used to work at heights, trenches and confined places
18. safe working practices specific to working with electrical equipment and system e.g. lock out/tag out, ptw, etc.



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19. preventative and remedial actions to be taken in the case of exposure to toxic materialstoxic materials: solvents, flux, lead exposure: ingested, contact with skin, inhaledpreventative action: ventilation, masks, protective clothing/ equipment); remedial action: immediate first aid, report to supervisor
20. importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
21. precautionary measures taken to prevent fire accident
22. various causes of firecauses of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.
23. correct technique(s) of using a fire extinguisher
24. different methods of extinguishing fire
25. different materials used for extinguishing firematerials: sand, water, foam, co2, dry powder
26. emergency rescue techniques applied during a fire hazard
27. various types of safety signs and what they mean
28. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
29. rescue techniques applied during fire hazards
30. good housekeeping in order to prevent fire hazards
31. correct use of a fire extinguisher
32. how to free a person from electrocution
33. basic techniques of bandaging
34. perform artificial respiration and the cpr process

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write an accident/ incident report in local language and/or english
2. read and comprehend basic content to read labels, charts, signages
3. read and comprehend basic english to read manuals of operation
4. read an accident/ incident report in local language or english
5. question coworkers appropriately in order to clarify instructions and other issues
6. give clear instructions to coworkers and subordinates
7. remain congenial while discussing and debating issues with co-workers
8. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
9. ask for, provide and receive required assistance where possible to ensure achievement of work-related objectives
10. follow organizational rule-based decision-making process
11. take decisions with systematic course of actions and/or response
12. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
13. build customer relationships and use customer centric approach



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14. think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
15. seek appropriate assistance from other sources to resolve problems
16. report problems that cannot be resolved to the appropriate authority
17. identify cause and effect relations in own area of work
18. work systematically and logically to resolve the issues, identify the cause and anticipate unexpected results
19. critically evaluate the organizational processes to improve them

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	19	43	-	-
PC1. • use protective clothing/equipment for specific tasks and work conditions Protective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visor equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator	1	2	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace Hazards-electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions	1	2	-	-
PC3. follow procedures for safe working with electrical equipment Safe Working Procedures: tag out/lock out, ptw (permit to work)	1	2	-	-
PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc	1	2	-	-
PC5. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC6. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC7. ensure positive isolation of electrical equipment & system as per the given standards	1	1	-	-
PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installed Parameters: temperature, pressure, flow and current	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.	1	2	-	-
PC10. use various methods of accident prevention in the work environment Methods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors	1	2	-	-
PC11. identify and retrieve general health and safety equipment in the workplace General Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.	1	2	-	-
PC12. set up and use scaffolds, elevated platforms and ladders safely Set up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc	1	3	-	-
PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify them Faults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/unfixed nuts or bolts, etc.	1	3	-	-
PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa	-	3	-	-
PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage	1	3	-	-
PC16. store flammable materials and machine lubricating oil safely and correctly	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	2	-	-
PC18. use good housekeeping practices at all times Good Housekeeping Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces	1	2	-	-
PC19. identify common hazard signs displayed in various areas Various Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.	1	2	-	-
PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	2	-	-
<i>Follow fire safety practices</i>	3	8	-	-
PC21. use various appropriate fire extinguishers for different types of fires correctly Types of Fires: class a, b, c, d and e	1	2	-	-
PC22. use appropriate rescue techniques during fire hazard	1	2	-	-
PC23. maintain good housekeeping in order to prevent fire hazards	1	2	-	-
PC24. use a fire extinguisher correctly in case of fire	-	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	19	-	-
PC25. administer appropriate first aid to the victims for the relevant injuries Injuries: bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments	1	4	-	-
PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergencies Emergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC30. free an affected person from electrocution safely as per the recommended procedure	1	2	-	-
PC31. move injured people and others to a safe place during emergencies	-	3	-	-
PC32. write an incident report or dictate a report to another person, and send report to the authorized person Incident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified	2	2	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Distribution, Transmission, Generation, Distribution Downstream
Occupation	Operation & Maintenance - Generation Supervision - Distribution Erection, Installation and Commissioning - Distribution Surveying - Transmission Operation & Maintenance - Transmission
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	11/02/2023
NSQC Clearance Date	22/08/2019



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PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



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- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



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PSS/N2503: Prepare for primary and secondary system of grid substation erection

Description

This unit is about general system layout design of primary and secondary system suitable for electrical environment present in high volt substation 66/33/11 kV following the latest engineering practice and ensuring long term compatibility requirement and continuity of equipment supply on a concept of bay oriented, distributed intelligence (SCADA) and safety of operating staff.

Scope

This unit/ task covers the following:

- Identify requirements and condition of equipment
- Perform preparatory tasks

Elements and Performance Criteria

Identify requirements and condition of equipment

To be competent, the user/individual on the job must be able to:

- PC1.** read and correctly interpret substation layout drawings
- PC2.** identify infrastructure present in the substation such as grid station, number of bays, number of incoming and outgoing feeders etc.
- PC3.** inspect loading and unloading of packed equipment at site, opening of accessories from packages to ensure there is no damage
- PC4.** identify manufacturing defects like defective design, poor material quality, poor workmanship and poor packing of equipment/transformer for transportation/shifting

Perform preparatory tasks

To be competent, the user/individual on the job must be able to:

- PC5.** carry out pipe and plate earthing and ensure that the earth resistance of each earth pit is marked with date of installation and next due date
- PC6.** fix and grout foundation bolts
- PC7.** lay gravel in the substation yard
- PC8.** ensure that the yard fencing, switchyard illumination and luminaries installation are done as per correct procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant standards, working procedures and policies of organization
2. main purpose and object of organization
3. organisational department and reporting structures
4. conditions and terms of own employment



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5. own job role and responsibilities
6. layout of the facility and the work area
7. organizational safety and security guidelines and procedures
8. principles of electricity
9. power system: how power flows, generation, transmission and distribution number of bays, number of incoming and outgoing feeders, load management through single or double bus, substation network, ring system, back feed etc.
10. line components, towers, poles, single circuit, double circuit, overhead, underground conductors and cables
11. gantry structure, structure lay out, types of porcelain insulators, overhead conductors, clamps used in station.
12. outdoor and indoor equipment like power transformer, circuit breaker, isolator, las, station transformers, cts, pts, and cvts
13. dc distribution board (dcbd), ac distribution board (acdb) and control cable and circuitry
14. functioning of lightening arrestors (la)
15. importance of proficiency in measurement of layout dimensions, grouting bolt, holes of fitting, cables for laying etc.
16. types of earthing used in grid station, its significance, why earth connection with each equipment is required
17. how to measure earth resistance and earthing switch.
18. use of complete tools, tackles and safety gadgets required in grid station erection and commissioning
19. safety precautions as per safety manual
20. importance of maintaining records and documentation of all equipment like name plate, pre-commission test report and manuals
21. importance of reporting problem to appropriate authorities appropriate authorities: eg. junior engineer, officer in charge
22. hazards associated with carrying out substation construction and installation process and how they can be minimized
23. importance and procedures to maintain cleanliness of working area, and 5s

Generic Skills (GS)

User/individual on the job needs to know how to:

1. communicate effectively in writing as per requirement of site work
2. write the information communicated by the engineer or in-charge of work
3. write properly about the technical problems and other conditions of site
4. note down observations, critical points and location of site related work
5. write using the followed metric system for all measurements
6. read and correctly interpret written sentences and paragraphs in work related documents
7. read and correctly interpret the rules and methods stated in the organizational standard operating procedure documents



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8. read equipment manuals to correctly interpret the equipment operation and process requirements
9. discuss task lists, schedules and activities with the concerned authority
10. communicate clearly with the team/group members
11. listen to the information given by the junior engineer and clarify doubts by asking relevant questions
12. make appropriate work-related judgments based on policy and procedures
13. identify complex problems and review related information to develop and evaluate solutions
14. follow organizations rule-based decision-making process
15. plan and organise tasks to meet deadlines
16. plan in advance the maintenance schedule daily, weekly and monthly
17. organize manpower, tools and kits for testing and maintenance
18. organize the shutdown of equipment in a planned manner
19. plan maintenance work to ensure it is completed within specified time
20. build customer relationships and use customer centric approach in taking decisions
21. identify problems and review related information to develop and evaluate options and implement solutions
22. monitor problems to take corrective action with respect to individuals and organizations
23. analyse problems and changes in conditions, operations, and the environment to solve problems
24. analyse the problems in the equipment to identify root causes
25. collect information and technical data and define process for equipment testing
26. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
27. develop holistic and comprehensive profile of erection of hv (high voltage) equipment based on segregated and discrete process stages
28. critically evaluate operation parameters in relation to grid station features intended to ensure a match



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify requirements and condition of equipment</i>	16	35	-	-
PC1. read and correctly interpret substation layout drawings	5	8	-	-
PC2. identify infrastructure present in the substation such as grid station, number of bays, number of incoming and outgoing feeders etc.	4	10	-	-
PC3. inspect loading and unloading of packed equipment at site, opening of accessories from packages to ensure there is no damage	4	9	-	-
PC4. identify manufacturing defects like defective design, poor material quality, poor workmanship and poor packing of equipment/transformer for transportation/shifting	3	8	-	-
<i>Perform preparatory tasks</i>	14	35	-	-
PC5. carry out pipe and plate earthing and ensure that the earth resistance of each earth pit is marked with date of installation and next due date	5	10	-	-
PC6. fix and grout foundation bolts	3	8	-	-
PC7. lay gravel in the substation yard	3	9	-	-
PC8. ensure that the yard fencing, switchyard illumination and luminaries installation are done as per correct procedure	3	8	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2503
NOS Name	Prepare for primary and secondary system of grid substation erection
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/03/2019
Next Review Date	11/03/2023
NSQC Clearance Date	22/08/2019



Qualification Pack

PSS/N2504: Erect and install substation equipment

Description

This unit covers the competencies required to erect high volt substation 66/33/11 kV following the latest engineering practice and ensuring long term compatibility requirement and continuity of equipment supply on a concept of bay oriented, distributed intelligence (Supervisory Control and Data Acquisition or SCADA). It also covers the respective health and safety competencies require to perform such operation.

Scope

This unit/task covers the following:

- Erect and commission outdoor and indoor equipment in 66kV or 33kV Grid Station
- Erect and commission the control and protection system in 66kV or 33 kV grid station

Elements and Performance Criteria

Erect and commission outdoor and indoor equipment in 66kV or 33kV Grid Station

To be competent, the user/individual on the job must be able to:

- PC1.** lay the power transformer on rails
- PC2.** check the alignment of the power transformer as per norms
- PC3.** check the external porcelain of the bushings for cracks and/or contamination
- PC4.** ensure secure connections at hv (high voltage) and lv (low voltage) bushings
- PC5.** ensure the optimum oil level is maintained in the transformer
- PC6.** fit auxiliary cooling system like cooling fans, pumps, oil coolers
- PC7.** install the circuit breaker as per the equipment manual/ recommendation of the manufacturer
- PC8.** install current transformers (cts), potential transformers (pts), capacitor voltage transformers (cvts) and lightning arrestors (las) as per the equipment manual/ recommendation of manufacture
- PC9.** earth the equipment appropriately with earth connection as per requirement

Erect and commission the control and protection system in 66kV or 33kV grid station

To be competent, the user/individual on the job must be able to:

- PC10.** ensure that all switch gear and control panels are properly aligned and grouted
- PC11.** ensure back door of control panel is marked with name of feeder, and is numbered, approachable, clean, dust free and vermin protected
- PC12.** check status of overcurrent (o/c) and earth fault (e/f) relays along with their settings, flag, etc. to avoid cut-offs
- PC13.** install dc distribution board (dcdB) and ac distribution board (acdB)
- PC14.** lay control cables
- PC15.** check the status of hrc (high rupturing capacity) fuse to assess need for replacement
- PC16.** check interlocks



Qualification Pack

PC17. mount ict (inter connecting transformer), auxiliary relays and other accessories as per specified norms

PC18. install battery bank and charging panel as per standards

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant standards, working procedures and policies of organization
2. main purpose and object of organization
3. organisational departments and reporting structures
4. conditions and terms of own employment
5. own job role and responsibilities
6. layout of the facility and the work area
7. organizational safety and security guidelines and procedures
8. principles of electricity
9. different types of power systems: power system: how power flows, generation, transmission and distribution number of bays, number of incoming and outgoing feeders, load management through single or double bus, substation network, ring system, back feed etc.
10. line components towers, poles, single circuit, double circuit, overhead, underground conductors and cables
11. gantry structure, structure lay out, types of porcelain insulators, overhead conductor and aa conductors, clamps used in station
12. operating principle of power transformer, its main component, auxiliary components and accessories
13. difference between power and distribution transformer
14. oltc (on load tap changer) and its mechanism
15. operating principle of switch gears (circuit breakers) and how it operates under fault current
16. benefits of operating medium of various types of circuit breaker types: oil circuit breakers (ocb), minimum oil circuit breakers (mocb), air circuit breakers (acb), sf, vacuum circuit breakers
17. impact of fault current due to short circuit, over current and earth fault and elements of the related protection system
18. working of instrument transformers like cts, pts, and cvts
19. control panel and its in-built measuring instruments, accessories like heater, lamp, door switch, hrc fuse, relays, auxiliary, icts etc.
20. use of battery panel, battery bank, trickle charging, battery status, electrolyte level, specific gravity of electrolyte, safety measures in repair and maintenance, ventilation etc.
21. elements of dc distribution board (dcdb), ac distribution board (acdb), control cable and related circuitry
22. types of earthing used in grid station, its significance, why earth connection is required with each equipment, measurement of earth resistance and earthing switch
23. types of lightening arrestors (las) like rod gap, horn gap, expulsion and metal oxide arrestor



Qualification Pack

24. elements of a shunt capacitor bank, its function to improve pf, switching operation, repair, replacement and maintenance
25. how to operate functions of transformer oil filter machine, inlet and outlet connection, how it purifies and clean from chemical impurities, slug, carbon and moisture
26. how to use complete tools, tackles and safety gadgets required in grid station erection and commissioning
27. use of safety, t and p, and testing equipment: hydraulic crimping kit, tool box kit and safety gadgets like discharge rod, earth chain, neon tester, signage, danger notice, cordon tape etc.
28. application of approved maintenance procedures and regulation
29. how to take safety precautions as per safety manual

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write official notes
2. write the information communicated by the work in-charge
3. write accurately about the technical problems and other conditions of site
4. note down observations, critical points and location of site related work
5. fill in all technical forms and data as per guidelines and format
6. read and correctly interpret written sentences and paragraphs
7. read and correctly interpret units of metric system for all measurements
8. read and correctly interpret the process required for performing work
9. read and correctly interpret the rules and methods from policy and procedure documents
10. read and correctly interpret equipment manuals to identify operational details and specifications
11. discuss task lists, schedules and activities with the concerned authority
12. effectively communicate with the team/group members
13. listen and correctly interpret the information given by superiors
14. communicate clearly with the team and other staff
15. make appropriate work-related judgments based on policy and procedures
16. identify complex problems and review related information to develop and evaluate solutions
17. follow organizations rule-based decision-making process
18. plan and organize tasks to meet deadlines
19. organize the shutdown of equipment in a planned manner
20. plan and organize tasks to meet quantity and quality specifications and targets
21. build customer relationships and use customer centric approach in taking decisions
22. identify problems and review related information to develop and evaluate options and implement solutions
23. monitor problems to take corrective action with individuals and organizations
24. analyse problems and changes in conditions, operations, and the environment to solve problems



Qualification Pack

25. analyse the problems in the equipment to identify root causes
26. collect information and technical data and define process for equipment testing
27. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
28. critically evaluate operation parameters in relation to grid stations intended features in order to ensure a match
29. develop a holistic and comprehensive profile for erection of hv equipment



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Erect and commission outdoor and indoor equipment in 66kV or 33kV Grid Station</i>	15	34	-	-
PC1. lay the power transformer on rails	2	3	-	-
PC2. check the alignment of the power transformer as per norms	2	3	-	-
PC3. check the external porcelain of the bushings for cracks and/or contamination	2	3	-	-
PC4. ensure secure connections at hv (high voltage) and lv (low voltage) bushings	1	3	-	-
PC5. ensure the optimum oil level is maintained in the transformer	1	4	-	-
PC6. fit auxiliary cooling system like cooling fans, pumps, oil coolers	1	3	-	-
PC7. install the circuit breaker as per the equipment manual/ recommendation of the manufacturer	2	5	-	-
PC8. install current transformers (cts), potential transformers (pts), capacitor voltage transformers (cvts) and lightning arrestors (las) as per the equipment manual/ recommendation of manufacture	2	5	-	-
PC9. earth the equipment appropriately with earth connection as per requirement	2	5	-	-
<i>Erect and commission the control and protection system in 66kV or 33kV grid station</i>	15	36	-	-
PC10. ensure that all switch gear and control panels are properly aligned and grouted	1	3	-	-
PC11. ensure back door of control panel is marked with name of feeder, and is numbered, approachable, clean, dust free and vermin protected	1	4	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check status of overcurrent (o/c) and earth fault (e/f) relays along with their settings, flag, etc. to avoid cut-offs	2	3	-	-
PC13. install dc distribution board (dcdb) and ac distribution board (acdb)	2	6	-	-
PC14. lay control cables	1	4	-	-
PC15. check the status of hrc (high rupturing capacity) fuse to assess need for replacement	2	4	-	-
PC16. check interlocks	2	3	-	-
PC17. mount ict (inter connecting transformer), auxiliary relays and other accessories as per specified norms	2	4	-	-
PC18. install battery bank and charging panel as per standards	2	5	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2504
NOS Name	Erect and install substation equipment
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/03/2019
Next Review Date	11/03/2023
NSQC Clearance Date	22/08/2019



Qualification Pack

PSS/N2505: Perform testing, commissioning and energization of substation equipment

Description

This unit is about commissioning and energizing various switchgear equipment installed in the control room and outdoor switchyard after thorough testing and clearance under the supervision.

Scope

This unit/task covers the following:

- Perform testing, commissioning and energization

Elements and Performance Criteria

Perform testing, commissioning and energization of substation equipment

To be competent, the user/individual on the job must be able to:

- PC1.** prepare and maintain records of installation, commissioning, tests, results and energization
- PC2.** check the proper working of oltc (on load tap changer) and various gauges of the transformer
- PC3.** check the functioning of alarm circuits and trip circuit of the transformer is as per specified norms
- PC4.** check the compressed air pressure in the transformer is as per specifications
- PC5.** check pneumatic drive as per specifications
- PC6.** check the sf gas pressure as per standards
- PC7.** check for any oil and hydraulic fluid leakages
- PC8.** perform various tests on transformer such as ir test, tan test, ratio test etc. and compare the same with the data obtained from manufacturer
- PC9.** test tripping mechanism for conditions such as short circuit, over current and earth fault
- PC10.** check the functioning of circuit breaker status indicator as per norms
- PC11.** lubricate all moving parts/ assembly by applying petroleum jelly to all contacts
- PC12.** check the status indicator of bushings of ct, pt, cvt is as per norms
- PC13.** check the earth connection of secondary circuit is correctly done
- PC14.** mark the earth resistance of each earth pit with installation date and the next due date
- PC15.** provide double earth connection for each equipment
- PC16.** check the availability and standard phase difference in the currents of each phase of the capacitor row
- PC17.** check the lightning arrestor (la) stacks for any cracks
- PC18.** determine the ir value (insulation resistance) of lightning arrestor (la)
- PC19.** check the tightness of t clamps and pg clamps (parallel groove clamp)
- PC20.** check proper functioning of the rtu (remote terminal unit) and communication panel for scada (supervisory control and data acquisition) system
- PC21.** check the test terminal block for availability of secondary current and voltage



Qualification Pack

PC22. maintain the electrolyte level of each cell with addition of distilled water

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant standards, working procedures and policies of organization
2. main purpose and object of organization
3. department and reporting structure
4. conditions and terms of own employment
5. own job role and responsibilities
6. safe working practices followed in the organisation
7. principles of electricity
8. elements of the power system: how power flows, generation, transmission and distribution number of bays, number of incoming and outgoing feeders, load management through single or double bus, substation network, ring system, back feed etc.
9. line components towers, poles, single circuit, double circuit, overhead, underground conductors and cables
10. gantry structure, structure lay out, types of porcelain insulators, overhead conductor and accessories, clamps used in station
11. operating principle of power transformer, its main components, auxiliary components and accessories.
12. difference between power and distribution transformer
13. operating principle of switchgears (cb), how it operates under fault current, benefits of operating medium of ocb, mccb, acb, sf, vacuum circuit breakers
14. the impact of fault current due to short circuit, over current and earth fault and the relevant protection systems requirement
15. instrument transformers like cts, pts, and cvts
16. control panel and its in built measuring instruments, accessories like heater, lamp, door switch, hrc fuse, relays, auxiliary, icts etc.
17. use of battery panel, battery bank, trickle charging, battery status, electrolyte level, specific gravity of electrolyte, safety measures in repair and maintenance, ventilation etc.
18. dc distribution board (dcdb), ac distribution board (acdb) and control cable and circuitry
19. tripping mechanism short circuit, earth fault, over current, low frequency etc. control panel relay for 66 or 33 kv outdoor switch yard
20. rtu and communication panel for scada
21. key faults occurred in major equipment like power transformer, distribution transformer, circuit breakers, control and relay panels etc., their causes and protection system to avoid break down.
22. functioning of lightening arrestors (la).
23. types of las like rod gap, horn gap, expulsion and metal oxide arrestor.
24. types of earthing used in grid station, its significance, why earth connection with each equipment, measurement of earth resistance and earthing switch.



Qualification Pack

25. elements of shunt capacitor bank, its function to improve pf, switching operation, repair, replacement and maintenance.
26. operating functions of transformer oil filter machine, inlet and outlet connection, how it purifies and cleans from chemical impurities, slug, carbon and moisture
27. complete tools, tackles and safety gadgets required in grid station erection and commissioning
28. use of safety, t and p, and testing equipment: multimeter, partial discharging kit, circuit breakers testing kit, meggar, sf6 gas leakage kit, digital tong tester, oil testing kit, battery testing kit, hydraulic crimping kit high voltage testing kit, protection testing kit, fault locator kit, tool box kit and safety gadgets like discharge rod, earth chain, neon tester, signage, danger notice, cordon tape etc
29. how to take safety precautions as per safety manual
30. importance of maintaining records and documentation of all equipment like name plate, pre-commission test report and manuals
31. cleanness of working area, and how to maintain 5s

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write official notes
2. write the information communicated by the work in-charge
3. write properly about the technical problems and other conditions of the site
4. note critical points and testing repair observations
5. write about the condition of equipment
6. fill in all technical forms and data as per guidelines and format
7. read and correctly interpret written sentences and paragraphs
8. read and correctly interpret units of metric system for all measurements
9. read and correctly interpret the process required for performing work
10. read and correctly interpret the rules and methods from policy and procedure documents
11. read and correctly interpret equipment manuals to identify operational details and specifications
12. discuss task lists, schedules and activities with the engineer
13. effectively communicate with the team/group members
14. interact with the junior engineer to clarify instructions and doubts
15. communicate clearly with the team and other staff
16. make appropriate work-related judgments based on policy and procedures
17. identify complex problems and review related information to develop and evaluate solutions
18. follow organizations rule-based decision-making process
19. plan and organise tasks to meet deadlines
20. plan in order to organize maintenance schedule daily, weekly and monthly
21. organize for the required manpower, tools and kits for performing testing and maintenance of equipment
22. organize the shutdown of equipment in a planned manner



Qualification Pack

23. plan work of maintenance completed within specified time
24. build customer relationships and use customer centric approach in taking decisions
25. identify problems and review related information to develop and evaluate options and implement solutions
26. monitor problem solving to take corrective action with individuals and organizations
27. analyse problems and changes in conditions, operations, and the environment to solve problems
28. analyse the problems in the equipment to identify root causes
29. collect information and technical data and define process for equipment testing
30. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
31. develop holistic and comprehensive profile of products based on segregated discrete process stages
32. critically evaluate operation parameters in relation to product features intended to ensure a match

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform testing, commissioning and energization of substation equipment</i>	30	70	-	-
PC1. prepare and maintain records of installation, commissioning, tests, results and energization	3	3	-	-
PC2. check the proper working of oltc (on load tap changer) and various gauges of the transformer	2	3	-	-
PC3. check the functioning of alarm circuits and trip circuit of the transformer is as per specified norms	1	3	-	-
PC4. check the compressed air pressure in the transformer is as per specifications	1	3	-	-
PC5. check pneumatic drive as per specifications	1	3	-	-
PC6. check the sf gas pressure as per standards	1	3	-	-
PC7. check for any oil and hydraulic fluid leakages	1	3	-	-
PC8. perform various tests on transformer such as ir test, tan test, ratio test etc. and compare the same with the data obtained from manufacturer	2	4	-	-
PC9. test tripping mechanism for conditions such as short circuit, over current and earth fault	2	4	-	-
PC10. check the functioning of circuit breaker status indicator as per norms	1	3	-	-
PC11. lubricate all moving parts/ assembly by applying petroleum jelly to all contacts	1	4	-	-
PC12. check the status indicator of bushings of ct, pt, cvt is as per norms	1	3	-	-
PC13. check the earth connection of secondary circuit is correctly done	1	3	-	-
PC14. mark the earth resistance of each earth pit with installation date and the next due date	1	4	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. provide double earth connection for each equipment	1	3	-	-
PC16. check the availability and standard phase difference in the currents of each phase of the capacitor row	1	2	-	-
PC17. check the lightning arrestor (la) stacks for any cracks	1	3	-	-
PC18. determine the ir value (insulation resistance) of lightning arrestor (la)	2	4	-	-
PC19. check the tightness of t clamps and pg clamps (parallel groove clamp)	1	3	-	-
PC20. check proper functioning of the rtu (remote terminal unit) and communication panel for scada (supervisory control and data acquisition) system	1	3	-	-
PC21. check the test terminal block for availability of secondary current and voltage	2	2	-	-
PC22. maintain the electrolyte level of each cell with addition of distilled water	2	4	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2505
NOS Name	Perform testing, commissioning and energization of substation equipment
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/03/2019
Next Review Date	11/03/2023
NSQC Clearance Date	22/08/2019

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- 1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70



Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	30	70	-	-	100	15
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2503.Prepare for primary and secondary system of grid substation erection	30	70	-	-	100	30
PSS/N2504.Erect and install substation equipment	30	70	-	-	100	30
PSS/N2505.Perform testing, commissioning and energization of substation equipment	30	70	-	-	100	20
Total	150	350	-	-	500	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
HV	High Voltage
CPR	Cardiopulmonary Resuscitation
PPE	Personal Protective Equipment
SCADA	Supervisory Control And Data Acquisition
CT	Current Transformers
PT	Potential Transformers
CVT	Capacitor Voltage Transformers
DCDB	Dc Distribution Board
ACDB	Ac Distribution Board
RTU	Remote Terminal Unit
LA	Lightening Arrestor
kV	Kilo Volt
O/C relay	Overcurrent Relay
E/F relay	Earth Fault Relay
HRC	High Rupturing Capacity
ICT	Inter-Connecting Transformer
T/R	Transformer
OLTC	On Load Tap Changer
CB	Circuit Breaker
OCB	Oil Circuit Breakers
MOCB	Minimum Oil Circuit Breakers



Qualification Pack

ACB	Air Circuit Breakers
IR test	Insulation Resistance Test
IR value	Insulation Resistance
PG clamp	Parallel Groove Clamp
SF	Sulfur Hexafluoride
CO2	Carbon Di-Oxide

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.